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Case 24-2023: A 43-Year-Old Man with a Pulmonary Nodule

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PRESENTATION OF CASE

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Dr. Margo A. Peyton (Medicine): A 43-year-old man was evaluated in the multidisciplinary Pulmonary Nodule Clinic of this hospital because of an incidentally found pulmonary nodule.

One month before the current evaluation, the patient, who lived in Panama, was seen at a primary care clinic in Colombia for a routine annual physical examination. He was in his usual state of good health, but because of a family history of cancer, he requested that imaging studies be obtained.

Dr. Jo-Anne O. Shepard: Computed tomography (CT) of the neck, chest, and abdomen (Fig. 1), performed after the administration of intravenous contrast material, revealed a pulmonary nodule (measuring 12 mm by 13 mm) in the right middle lobe.

Dr. Peyton: During the next month, follow-up testing was performed. The 5-hydroxy-indoleacetic acid level in a 24-hour urine specimen and the chromogranin A level in blood were reportedly normal, as were the results of pulmonary function testing and colonoscopy.

Dr. Shepard: ¹⁸F-fluorodeoxyglucose (FDG)–positron-emission tomography (PET) and CT (Fig. 2) revealed FDG uptake in the right-middle-lobe nodule (measuring 14 mm in diameter), with a maximum standardized uptake value of 3.8. There was also FDG uptake in multiple normal-sized right inferior paratracheal, subcarinal, and hilar lymph nodes. A right hilar lymph node (measuring 8 mm in diameter) had a maximum standardized uptake value of 6.7.

Dr. Peyton: The patient requested evaluation in the United States and was referred to the multidisciplinary Pulmonary Nodule Clinic of this hospital. On evaluation, the patient felt well. He had no fever, shortness of breath, cough, wheezing, hemoptysis, headache, vision changes, oral ulcers, nausea, vomiting, abdominal pain, arthralgias, or rash. He reported no recent illnesses.

The patient had a history of gastroesophageal reflux disease. Three years before this evaluation, chest radiography and abdominal ultrasonography, performed at his

request because of a family history of cancer, had reportedly shown no abnormalities. The patient took no medications and had no known drug allergies. He worked in the finance industry and had previously been a health care worker. He was born in northern South America and had moved to southern Central America 5 years earlier. He lived in an urban area with his wife, son, and dog.

The patient had visited nearly every country in North America, Central America, South America, and the Caribbean. Four years earlier, he had traveled by motorcycle from Alaska to Panama; 6 months earlier, he had traveled by motorcycle from Colombia to Chile. The patient had also traveled to several countries in western and eastern Europe, northern Africa, the Mediterranean, and the Pacific Islands. He had a physically active lifestyle that included weight lifting and bicycling at least four times per week. He had smoked two cigarettes daily and one cigar every 3 months for

2 years. His family history of cancer included breast cancer in his mother, prostate and pancreatic cancers in his father, cancer of unknown primary origin in his maternal grandfather, and ovarian cancer in his paternal grandmother.

On evaluation, the temporal temperature was 36.2°C, the blood pressure 133/82 mm Hg, the pulse 68 beats per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 27.1. He appeared well, and the lungs were clear on auscultation. There was no palpable cervical, supraclavicular, axillary, or inguinal lymphadenopathy. The white-cell count was 6970 per microliter (reference range, 4500 to 11,000), the hemoglobin level 12.7 g per deciliter (reference range, 13.5 to 17.5), and the platelet count 213,000 per microliter (reference range, 150,000 to 400,000). Results of kidney-function, liver-function, and coagulation tests were normal.

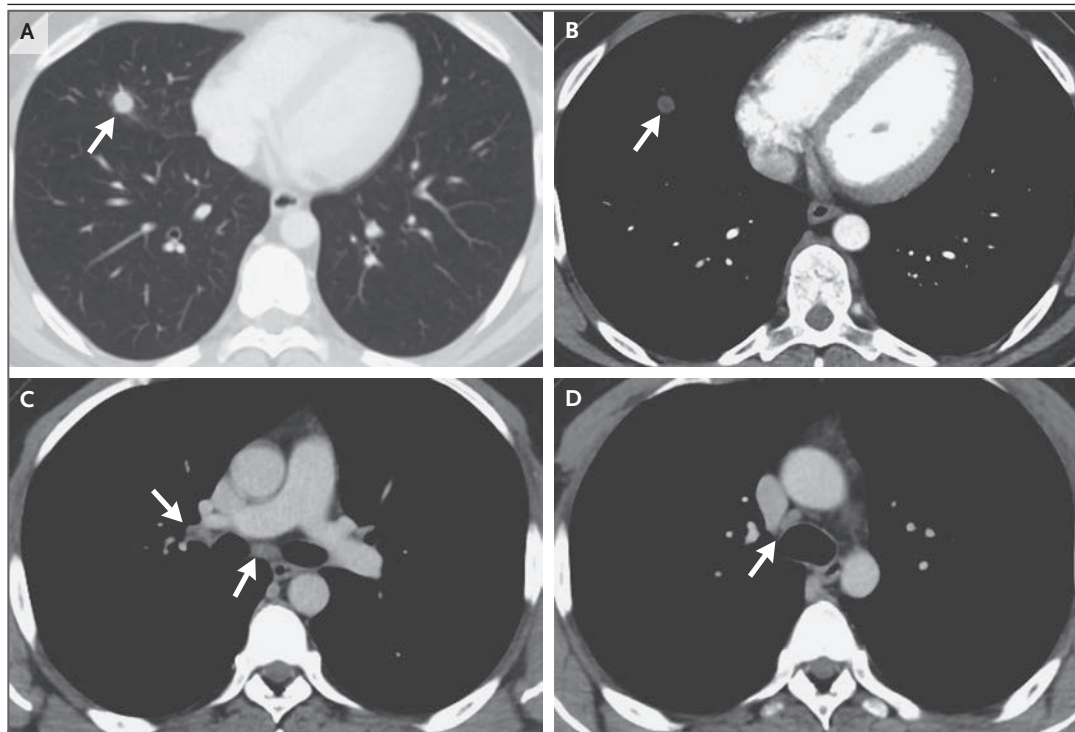


Figure 1. Chest CT.

Chest CT was performed after the administration of intravenous contrast material. Images obtained in lung windows show a 12-mm pulmonary nodule in the right middle lobe, abutting the major fissure (Panel A, arrow). The nodule has low attenuation and does not have calcification, macroscopic fat, or intense enhancement (Panel B, arrow). Normal-sized lymph nodes are present in the right hilum, subcarinal area, and right paratracheal region (Panels C and D, arrows).

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Angela J. Frank: Although I was not directly involved in this patient's care, I am aware of the final diagnosis in this case. When a pulmonary nodule measuring 6 mm or greater in diameter is either detected on screening or found incidentally, the patient is evaluated by a team in the Pulmonary Nodule Clinic at this hospital, which includes specialists from pulmonology, thoracic surgery, medical oncology, radiation oncology, interventional pulmonology, and interventional thoracic radiology, as well as nurse practitioners, an access nurse navigator, and a tobacco-treatment specialist.¹ In the Pulmonary Nodule Clinic, we consider salient aspects of the patient's history, as well as the characteristics of the nodule on imaging. In this case, notable aspects of the patient's history include his family history of cancer, former occu-

pation as a health care worker, residence in Central and South America, and extensive travel history. On the basis of the history and imaging findings, I will construct a differential diagnosis that focuses on possible inflammatory, neoplastic, and infectious causes of this patient's pulmonary nodule.

INFLAMMATORY CAUSES

The list of inflammatory conditions that can cause pulmonary nodules is extensive. It includes, but is not limited to, sarcoidosis, aspiration, IgG4-related disease, rheumatoid arthritis, vasculitis (e.g., granulomatosis with polyangiitis), amyloidosis, inflammatory bowel disease, and common variable immunodeficiency.

Of these diagnoses, sarcoidosis is the most likely in this case, on the basis of the presence of FDG-avid hilar lymph nodes. However, sarcoidosis is usually associated with smaller nodules that occur in a peribronchovascular and perifissural

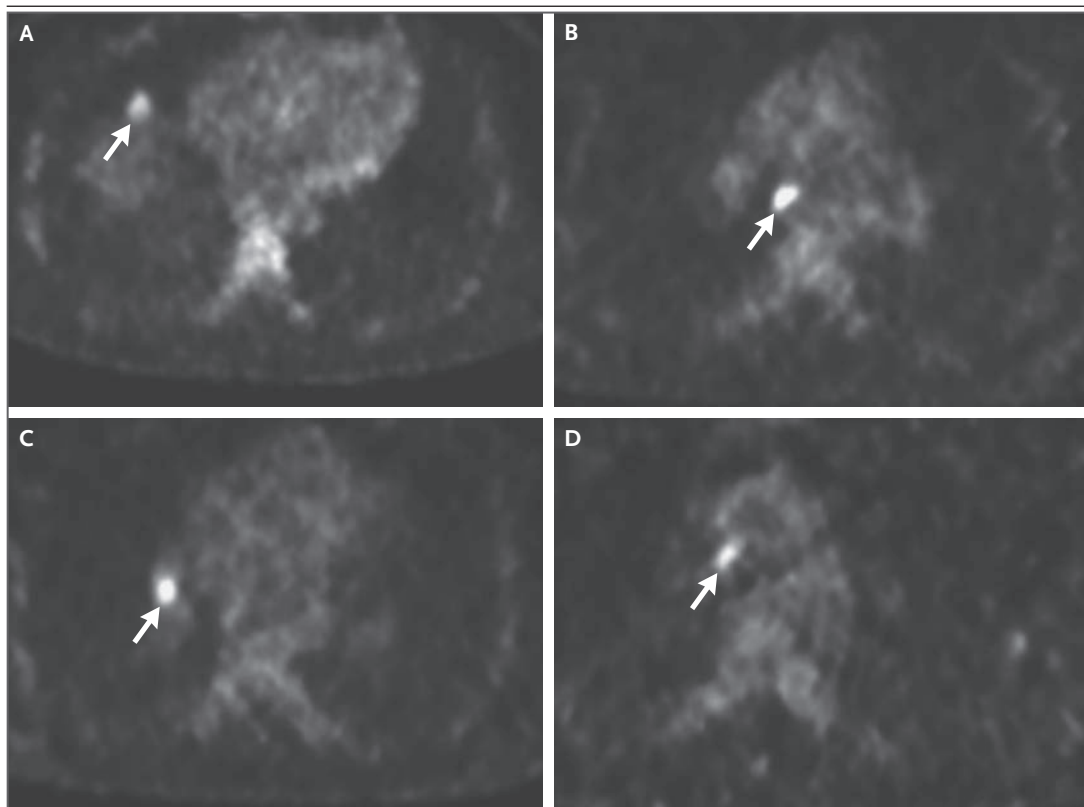


Figure 2. PET Scan.

¹⁸F-fluorodeoxyglucose (FDG)–positron-emission tomography (PET) was performed. Intense FDG uptake is present in the right-middle-lobe nodule (Panel A, arrow) and in right hilar and mediastinal lymph nodes (Panels B, C, and D; arrows).

distribution predominantly in the upper lobes. In addition, sarcoidosis is typically associated with bilateral, symmetric hilar lymphadenopathy. Although sarcoidosis can have an atypical presentation with a single large nodule, it is not the most likely diagnosis in this case overall.²

BENIGN LESIONS

A well-demarcated, round nodule in the lung is often a hamartoma, a benign tumor of fibrous tissue. The presence of fat or popcorn calcification in the nodule on CT is virtually diagnostic of a hamartoma, but the absence of this finding does not rule out the diagnosis.³ Hamartomas can grow very slowly but are not malignant and thus would not be associated with FDG-avid lymph nodes.

INFECTIOUS CAUSES

This patient's residence in Central America and extensive travel history are risk factors for many infections. Bacterial infections that can cause solitary pulmonary nodules, such as nocardiosis, are unusual in patients who are not immunocompromised and do not have underlying structural lung disease, so such processes are unlikely in this case.

Tuberculosis is a consideration, given the patient's former occupation as a health care worker and his residence in Central America. However, tuberculomas typically occur in the upper lobes, and other infections would be more common with his geographic exposures.⁴

Fungal infection is the most likely cause of this patient's pulmonary nodule. Cryptococcosis occurs worldwide, but disseminated disease is rare in immunocompetent patients. The infection is asymptomatic in approximately one third of immunocompetent patients.⁵ Pulmonary nodules that develop with cryptococcosis can be solitary or multiple but are usually peripherally located; mediastinal lymphadenopathy can occur but is uncommon.⁵

Blastomycosis can occur in immunocompetent patients and is frequently a subclinical infection. However, this patient had not visited regions where the fungus is endemic, such as the Mississippi River Valley or the Ohio River Valley, and the infection does not typically cause mediastinal or hilar lymphadenopathy.⁶

Coccidioidomycosis can be caused by one of two distinct species: *Coccidioides immitis*, which is endemic in California, and *C. posadasii*, which is endemic in the southwestern United States and

in Central and South America.⁷ The infection is asymptomatic in up to 60% of cases. Patients can present with self-limited pneumonia, but they frequently also have a solitary pulmonary nodule.⁸ As a result, the evaluation of lung nodules and lung cancer screening can be challenging in geographic regions where the disease is endemic.^{9,10}

Paracoccidioidomycosis is endemic in South America and can be acute or chronic. The chronic form has a long latency period, with symptoms or signs manifesting up to several decades after exposure.¹¹ Pulmonary nodules that develop with paracoccidioidomycosis can be either large and cavitary or small, randomly distributed, and irregularly shaped.¹²

Histoplasmosis is a fungal infection that is endemic in Central and South America.⁷ The disease was first described in a worker helping to construct the Panama Canal in 1906.¹³ Histoplasma is found at high concentrations in bird and bat droppings, so persons with exposure to chicken coops or caves are at an increased risk for infection. However, up to 90% of persons with exposure to the fungus have only mild or no symptoms.¹⁴

Histoplasmosis is associated with a range of pulmonary manifestations. Approximately 10% of patients present with acute pulmonary histoplasmosis, which includes pulmonary manifestations such as fever, cough, and dyspnea. Of these patients, 5% also have skin or joint manifestations, such as arthralgias, arthritis, erythema nodosum, or erythema multiforme.¹⁵ With acute pulmonary histoplasmosis, diffuse patchy opacities are seen on imaging; mediastinal and hilar lymphadenopathy are very common. Other pulmonary manifestations of histoplasmosis include chronic cavitary pulmonary histoplasmosis, which is seen mostly in older patients with chronic obstructive pulmonary disease, as well as mediastinal granulomas and mediastinal fibrosis.

Histoplasmosis is a very common cause of pulmonary nodules. In one case series focusing on infections that mimic lung cancer, histoplasmosis was the most common fungal cause.¹⁶ PET-CT imaging is not particularly helpful, given that the specificity of PET-CT for the evaluation of lung nodules in areas where histoplasma is endemic is approximately 61%.¹⁷ The diagnosis of histoplasmosis can be difficult in a patient with an isolated pulmonary nodule. Culture is unlikely to grow the organism, although narrow-based budding yeast forms can sometimes be seen on his-

topathological examination. Histoplasma antigen testing of urine, serum, or bronchoalveolar lavage fluid is effective in patients with disseminated histoplasmosis or acute pulmonary histoplasmosis but is unlikely to be helpful in this case. Antibody tests such as complement fixation or immunodiffusion assays could be helpful because they are often positive for years after the initial infection.¹⁵

On the basis of this patient's age, generally good health, and epidemiologic risk factors, I suspect that his pulmonary nodule is most likely due to an asymptomatic fungal infection such as histoplasmosis. However, I would consult an oncologist, a radiologist, and a thoracic surgeon to consider other diagnoses and next steps in the evaluation.

DR. ANGELA J. FRANK'S DIAGNOSIS

Pulmonary nodule most likely due to an endemic fungal infection.

ONCOLOGY EVALUATION

Dr. Inga T. Lennes: Incidental findings on chest CT are common. A chief concern with any incidental nodule is the possibility of cancer. In evaluating a nodule for possible cancer, it is important to consider the following factors: the size, the density (solid, subsolid, or mixed), the rate of change in size and density over time, the shape (round, ovoid, or irregular, with smooth, lobulated, or spiculated borders), the location (peripheral or central), and the patient's cancer risk (based on age, environmental exposures, and participation in screening). Neoplastic nodules can vary in histologic characteristics and in aggressiveness (from benign to malignant).

A premalignant neoplasm, such as atypical adenomatous hyperplasia, is typically small and subsolid. Carcinoma in situ — a group of non-invasive lesions that is a precursor to invasive adenocarcinoma or invasive squamous-cell carcinoma — is usually larger and subsolid or solid. The most common malignant pulmonary nodule is adenocarcinoma, which can be subsolid, mixed, or solid. Adenocarcinoma tends to grow more slowly than other neoplasms and therefore is more frequently identified incidentally. Other non-small-cell lung cancers, such as squamous-cell carcinoma, can be found incidentally. Such diag-

noses are difficult to distinguish on imaging and require a biopsy.¹⁸

Neuroendocrine lung cancers can be found incidentally on chest CT. They range from indolent lesions, such as typical carcinoid tumors, to aggressive cancers, such as small-cell lung cancer. Typical carcinoid tumors are usually spherical or ovoid and have well-defined borders that are sometimes lobulated. Atypical carcinoid tumors can look like typical carcinoid tumors or can have the appearance of traditional lung cancers, with an irregular shape and ill-defined, spiculated borders; multiple lesions and lymphadenopathy can be present. Large-cell neuroendocrine carcinoma of the lung is often indistinguishable from any non-small-cell lung cancer on imaging. The most aggressive neuroendocrine lung cancer is small-cell lung cancer, which does not usually manifest as a pulmonary nodule and classically manifests as a bulky mass with hilar lymphadenopathy.¹⁹

Metastatic lesions can be pulmonary nodules that are well-circumscribed, round, and peripherally located. The most common cancers that result in metastatic pulmonary nodules include lung cancer (originating in another part of the lung), colorectal cancer, renal carcinoma, pancreatic carcinoma, breast cancer, and testicular cancer. In addition, less common cancers that predominantly metastasize to the lung include choriocarcinoma, Ewing's sarcoma, melanoma, osteosarcoma, and thyroid cancer. To determine the best approach for obtaining a tissue diagnosis in this patient, the next step is to ask a radiologist and a thoracic surgeon to discuss options for biopsy.

RADIOLOGY EVALUATION

Dr. Shepard: Percutaneous needle biopsy of focal lesions of the lung, hilum, mediastinum, pleura, or chest wall can be performed with the use of CT guidance and with the patient under moderate sedation. As a rule, pulmonary nodules measuring at least 1 cm in diameter can be biopsied, provided that the needle can avoid large vessels and bronchi. In this patient, the right-middle-lobe nodule is a suitable target for percutaneous needle biopsy. However, the normal-sized hilar and mediastinal lymph nodes cannot be biopsied with this technique.

Needle aspiration or core biopsy could be per-

Figure 3. Lung-Biopsy Specimen.

Hematoxylin and eosin staining (Panel A) shows necrotizing granulomatous inflammation effacing the residual lymph node. At higher magnification (Panel B), palisading histiocytic inflammation with rare giant cells and central necrosis is shown. Grocott–Gomori methenamine silver special staining (Panel C) highlights numerous fungal forms that are round to ovoid, measure 3 to 4 μm in greatest dimension, and have focal narrow-based budding. The overall findings are consistent with histoplasmosis.

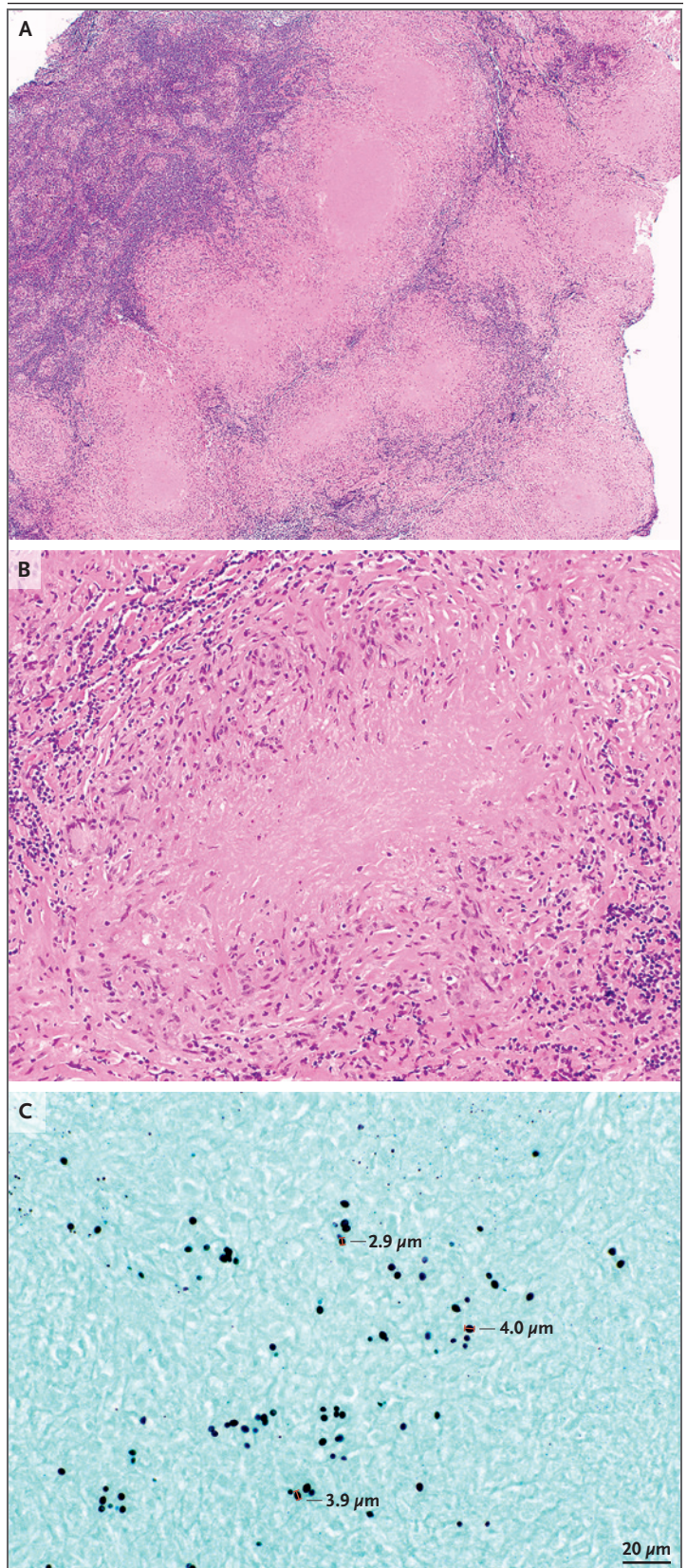
formed for cytologic and pathological examination, flow cytometry, molecular profiling, and microbiologic smear and culture. On-site cytopathological examination provides rapid feedback to increase the diagnostic yield and facilitate specimen handling. With percutaneous needle biopsy of pulmonary nodules, a sensitivity of 93 to 98% and a specificity of 98 to 100% have been reported for the diagnosis of cancer. The major complications include pneumothorax (10 to 20%), the need for placement of a chest tube (1 to 7%), and hemoptysis and bleeding (5%). Rare complications include air embolism, tumor seeding, and death.²⁰

THORACIC SURGERY EVALUATION

Dr. Lana Y. Schumacher: There are several diagnostic approaches to consider in the evaluation of this patient's pulmonary nodule and mediastinal lymph nodes. The first approach is endobronchial ultrasonography with transbronchial needle aspiration. Endobronchial ultrasonography allows for real-time visualization of mediastinal structures while transbronchial needle aspiration of a mediastinal lymph node is performed. This technique has a 91% sensitivity for establishing a diagnosis²¹ and is associated with a low risk of complications.²²

If transbronchial needle aspiration is unlikely to produce enough tissue for diagnosis, cervical mediastinoscopy can be performed. In this procedure, a small cervical incision is made, and a dissection plane is created in the pretracheal fascia. A rigid mediastinoscope is inserted, and mediastinal nodes can be dissected and biopsied. This technique has a high sensitivity (95.5%) and specificity (100%), with a diagnostic yield of 98.3%.²³

If the nodule is smaller and peripherally located, and it may not be within reach of a tradi-



tional bronchoscope, electromagnetic navigation bronchoscopy may be helpful in targeting the lesion for sampling. An electromagnetic generator with sensors, in combination with CT, allows for real-time visualization while a biopsy specimen of a lung nodule is obtained. The diagnostic yield is approximately 73% (77% sensitivity and 100% specificity).²⁴

With robotic bronchoscopy, a technique introduced in 2019, either electromagnetic navigation bronchoscopy or shape-sensing technology is used to guide a very small catheter (3.5 mm) out to the periphery of the lung. This small catheter can reach all 18 segments of the lung and allows for real-time visualization of the airway. For the assessment of small peripheral nodules, this technique has been associated with a diagnostic yield of higher than 80% and a low risk of complications.²⁵

This patient underwent endobronchial ultrasonography with transbronchial needle aspiration. To confirm the diagnosis, cervical mediastinoscopy was also performed.

PATHOLOGICAL DISCUSSION

Dr. Angela R. Shih: Histologic evaluation of a biopsy specimen obtained from the station 4R lymph node (Fig. 3A and 3B) showed palisading histiocytes with rare giant cells and large areas of central necrosis, findings consistent with necrotizing granulomatous inflammation. Grocott–Gomori methenamine silver special staining (Fig. 3C) highlighted numerous fungal forms within the areas of necrosis. The fungal forms were round to ovoid, measured 3 to 4 μm in greatest dimension, and at least occasionally appeared to have narrow-based budding. These morphologic findings are consistent with, but not diagnostic of, histoplasma, given that these characteristics can also be seen with cryptococcus. The absence of broad-based budding makes blastomyces unlikely. Fontana–Masson and mucicarmine staining both showed that the fungal organism had no capsule, a finding typical of histoplasma. Unfortunately, no specimen was available for microbiologic smear and culture.

Urinary histoplasma antigen testing was negative, as was serologic testing for cryptococcus antigen, histoplasma antigen, and histoplasma antibody. It is likely that these tests were negative because of the local, walled-off nature of the infectious process. However, a complement fixation assay was positive for histoplasma antibody to yeast antigen at a 1:64 dilution and was positive for histoplasma antibody to mycelial antigen at a 1:8 dilution.

Taken together, the overall morphologic and serologic findings were consistent with a diagnosis of histoplasmosis.

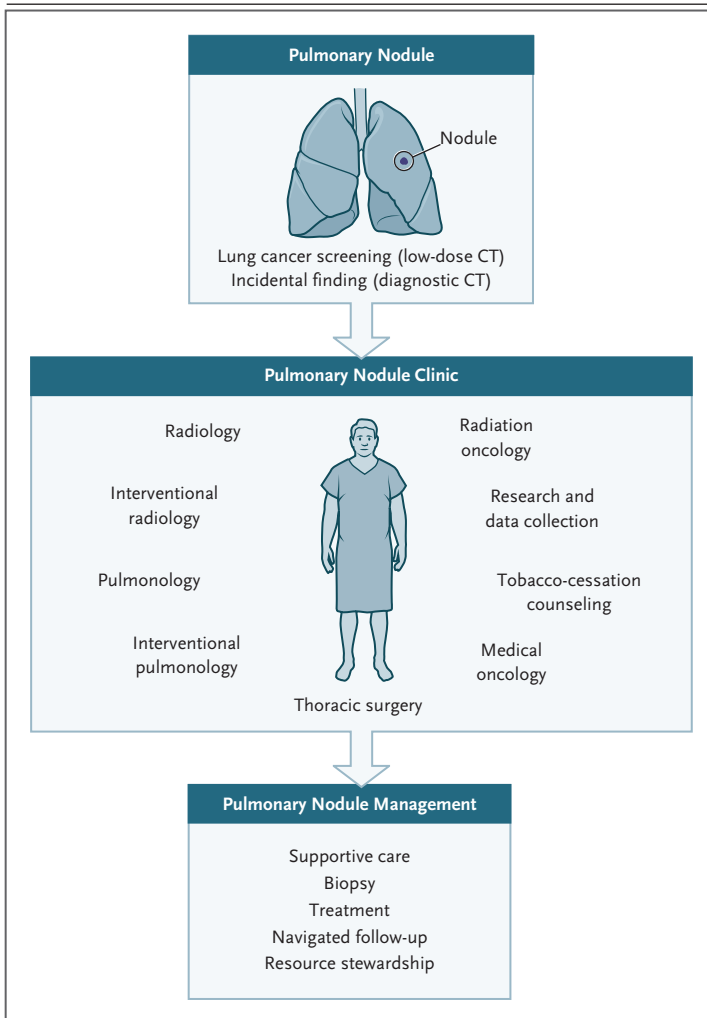


Figure 4. Structure and Patient Flow in the Multidisciplinary Pulmonary Nodule Clinic.

Management of pulmonary nodules involves input from specialists in several clinical disciplines. Patients with pulmonary nodules (≥ 6 mm) detected on screening or found incidentally are referred to the Pulmonary Nodule Clinic. Patients with pulmonary nodules may interact with various specialists, depending on features of the nodule and the patient. There are several options for the management of pulmonary nodules, depending on the size, location, and growth rate of the nodule, as well as the age, coexisting conditions, and preferences of the patient. Multidisciplinary discussion among a pulmonologist, thoracic surgeon, thoracic radiologist, and medical and radiation oncologists yields a unique treatment plan for each patient.

PATHOLOGICAL DIAGNOSIS

Histoplasmosis.

DISCUSSION OF MULTIDISCIPLINARY CARE

Dr. Lennes: This patient had undergone chest CT because of a family history of cancer. Although family history is part of many risk prediction models for lung cancer, it is important to review the current U.S. Preventive Services Task Force guidelines for lung cancer screening.²⁶ Current guidelines recommend annual lung cancer screening with low-dose CT in adults 50 to 80 years of age who have a 20-pack-year smoking history and currently smoke or have quit smoking within the past 15 years.^{26,27} Increasing numbers of patients are participating in lung cancer screening with low-dose CT, but most pulmonary nodules are still found incidentally, as was the case in this patient.

The management of pulmonary nodules is often suboptimal, with poor follow-up rates and a process that involves increasingly complex decisions. In 2012, the Pulmonary Nodule Clinic was founded at this hospital with the goal of improving the management of pulmonary nodules. The purpose of the clinic is to facilitate multidisciplinary discussion and efficient treatment of patients with high-risk pulmonary nodules. Patients are longitudinally followed by the multidisciplinary team until they have a definitive procedure or treatment, are eligible for yearly lung cancer screening ordered by their primary care team, or no longer need follow-up imaging be-

cause of nodule stability or age and coexisting conditions.

Each component of the Pulmonary Nodule Clinic (Fig. 4) contributes to providing an efficient and expedited opinion that draws from the expertise of multiple specialists. This case illustrates the need for multidisciplinary care and is representative of many of the cases seen in the clinic. As options for diagnosis and treatment expand and improve, the treatment of patients with pulmonary nodules will necessitate coordination and conversation among specialists. Multidisciplinary management of pulmonary nodules has been shown to promote guideline-concordant care and facilitate appropriate deviation from guidelines on the basis of expert input.²⁸

Dr. Schumacher: The patient was referred to the infectious disease clinic. After evaluation, no further treatment for histoplasmosis was recommended because the patient was asymptomatic and there was no evidence of active systemic disease. Our plan is to follow this patient for up to 2 years to ensure stability or continued regression of the pulmonary nodule.

Dr. Shepard: Chest CT performed 7 months after the initial evaluation revealed a decrease in the size of the right-middle-lobe nodule, from 14 mm to 8 mm in diameter.

FINAL DIAGNOSIS

Histoplasmosis.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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